Nithin Venkat Sharma P M

Coimbatore, Tamil Nadu
nithin.p.m2006@gmail.com | +91 73730 93900 | linkedin.com/in/nithin-p-m- | github.com/nithin0306

Summary

Computer Science undergraduate with full-stack development experience and a strong foundation in data structures and software design. Growing focus on backend engineering, system design, and AI-driven architectures, with hands-on exposure to building and integrating scalable applications. Quick to pick up new tools and frameworks, with a collaborative, detail-oriented approach to writing clean, maintainable code.

Education

Amrita Vishwa Vidyapeetham, Coimbatore Aug 2024 – May 2028
B.Tech in Computer Science and Engineering

- CGPA: 8.85 / 10

Experience

Web Developer, Tensor Club, Amrita Vishwa Vidyapeetham Jul 2025 – Apr 2026

- Built the 'Featured Projects' and 'All Projects' pages with React and Tailwind CSS, achieving a 99 Lighthouse performance score with 0.3s First Contentful Paint and zero cumulative layout shift.
- Implemented responsive, terminal-style interactive card layouts with uniform rendering across mobile and desktop viewports, contributing to 98 accessibility and 100 SEO scores for the pages.

Automation Intern, The Pedagogy Community Apr 2026 – Jun 2026

- Architected an AI pipeline that transforms raw webinar recordings into structured courses, reducing manual production time from 8 hours to under 15 minutes
- Bypassed API file-size constraints by building a multi-threaded chunking engine that runs 20-minute audio slices in parallel, maintaining perfect timestamp alignment
- Developed an automated speech-stitching and MCQ generation system that strips awkward Q&A pauses via FFmpeg and builds 5 scenario-based quizzes per lesson

Projects

Dayta | *React (Vite, TypeScript), FastAPI, Pinecone, Google Gemini API* Live

- Engineered an automated document ingestion system that processes multi-page academic schedules, generating high-density vector embeddings stored in a vector database for rapid semantic search
- Implemented conversational guardrails with windowed session memory, confining responses strictly to official university data to prevent hallucinated or out-of-scope answers across multi-turn conversations
- Built a lightweight, modular backend architecture utilizing FastAPI and asynchronous pipeline execution, ensuring end-to-end question answering handles multi-turn scheduling queries flawlessly

Archivist | *Python, LangGraph, GitHub Actions, LLMs, MCP* GitHub

- Engineered an autonomous GitHub Action using LangGraph and MCP to dynamically parse PR diffs, achieving 100% automated enforcement of configured Architecture Decision Records (ADRs)
- Enforced CI/CD governance via the GitHub API, catching and blocking injected violations such as client-side secret leaks during pipeline simulations
- Reduced the architectural review feedback loop from hours of manual inspection to under 60 seconds, maintaining development velocity with a secure webhook-driven override command

GoCDC (Event-Driven CDC Pipeline) | *Go, PostgreSQL, Apache Kafka, Redis, MongoDB, Docker* GitHub

- Engineered a Go-based Change Data Capture (CDC) engine utilizing PostgreSQL WAL logical decoding, streaming DML events to Redpanda at 1,400 events/sec to completely eliminate database polling overhead.
- Architected a fault-tolerant polyglot persistence pipeline routing data from Postgres to Redis and MongoDB via Kafka, achieving <150ms end-to-end replication lag and zero data loss during 500-concurrent-transaction benchmarks.
- Containerized the 6-node microservice ecosystem using multi-stage Docker builds, reducing Go binary image sizes to 20MB and orchestrated seamless local and production deployments via Docker Compose.

Technical Skills

- **Languages:** C, C++, Java, JavaScript, Python, SQL, Go (Golang)
- **Frameworks & Libraries:** React.js, FastAPI, Node.js, Express.js, LangChain, LangGraph
- **Databases:** PostgreSQL, MongoDB, Supabase
- **Tools & Technologies:** Git, GitHub, Docker, Postman, Linux
- **Core Concepts:** Data Structures, Problem Solving, Software Development Lifecycle (SDLC), Team Collaboration

Achievements & Open Source

- Secured 2nd place (out of 30+ teams) at Hack101 for developing SkillUp, an AI-powered resume analyzer that detects skill gaps and curates personalized project and course recommendations for target job roles
- Secured 3rd place (out of 80+ teams) at the Build2Break Hackathon for VITA-CARE, a voice-integrated, Gemini AI-powered microservice ecosystem that automates healthcare appointment scheduling and WhatsApp medication reminders
- Merged 4 pull requests across independent open-source repositories through GirlScript Summer of Code (GSSoC), spanning Redis-backed setup automation, API unit test coverage, and SEO metadata improvements